

Q.2. What is difference between primary key and foreign Key?



| Feature     | Primary Key                                                                     | <b>Foreign Key</b>                                                                                                     |
|-------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Definition  | Must contain unique values; no two records can have the same primary key value. | Can have duplicate values; multiple records in the child table can reference the same primary key in the parent table. |
| Uniqueness  | Must contain unique values; no two records can have the same primary key value. | Can have duplicate values; multiple records in the child table can reference the same primary key in the parent table. |
| Nullability | Cannot contain null values; every record must have                              | Can contain null values (unless explicitly                                                                             |

|          |                                                                                                |                                                                                                                 |
|----------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
|          | a valid primary key value.                                                                     | restricted), allowing for optional relationships.                                                               |
| Purpose  | To ensure each record is uniquely identifiable and maintain entity integrity within the table. | To establish a relationship between two tables and enforce referential integrity.                               |
| Location | Defined in the table being referenced (parent table).                                          | Defined in the table that references another table (child table).                                               |
| Example  | In a <b>Students</b> table, StudentID could be the primary key.                                | In a <b>Courses</b> table, StudentID could be a foreign key referencing StudentID in the <b>Students</b> table. |